Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Further Pure 1 – Further Differentiation

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Differentiate $\arcsin x$ from first principles or standard result. **[3]**

2. Differentiate $\arctan x$. **[2]**

3. Differentiate $\operatorname{arsinh} x$. **[2]**

4. Differentiate $\operatorname{arcosh} x$. **[2]**

5. Find $\dfrac{d}{dx}(\arcsin(x^2))$. **[3]**

6. Find $\dfrac{d}{dx}(x \arctan x)$. **[3]**

7. Use implicit differentiation to find $\dfrac{dy}{dx}$ if $x^2 + y^2 = 1$. **[3]**

8. Find $\dfrac{d^2y}{dx^2}$ if $x^2 + y^2 = 1$. **[4]**

9. Parametric: $x = \cos t$, $y = \sin t$. Find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$. **[4]**

10. Find the equation of the tangent to $y = \arcsin x$ at $x = \frac{1}{2}$. **[4]**

11. Differentiate $\ln(\arcsin x)$. **[3]**

12. Find $\dfrac{dy}{dx}$ if $y = x^{x}$. **[4]**

13. Differentiate $\arctan\left(\frac{1+x}{1-x}\right)$. **[4]**

14. Find the stationary points of $y = x \arcsin x$. **[4]**

15. State the formula for $\dfrac{d}{dx}(a^{x})$. **[2]**

Section B – Hard (Q16–25)

16. If $x = a\cosh t$, $y = b\sinh t$, find $\dfrac{dy}{dx}$ and the equation of the tangent at $t = t_0$. **[5]**

17. Find $\dfrac{d^2y}{dx^2}$ in terms of t for $x = t^2$, $y = t^3$. **[4]**

18. Differentiate $\operatorname{arsech} x$ and simplify. **[4]**

19. Use logarithmic differentiation to find $\dfrac{dy}{dx}$ if $y = \frac{(x+1)^2\sqrt{x-1}}{(x+2)^3}$. **[5]**

20. Find the curvature $\kappa = \frac{|y''|}{(1+y'^2)^{3/2}}$ for $y = \ln x$ at $x = 1$. **[5]**

21. Implicit: $x^3 + y^3 = 3xy$. Find $\dfrac{dy}{dx}$ at $(1, 1)$. **[4]**

22. Find the maximum value of $y = x^{1/x}$ for $x > 0$. **[5]**

23. Parametric circle $x = a\cos t$, $y = a\sin t$. Show that the tangent at t is $x\cos t + y\sin t = a$. **[4]**

24. Find $\dfrac{dy}{dx}$ and $\dfrac{d^2y}{dx^2}$ if $y = \arctan\left(\frac{2x}{1-x^2}\right)$. **[5]**

25. Differentiate $f(x) = \arcsin x + \arccos x$ and comment. **[3]**

End of Worksheet • *Show all working clearly.*